

Compact attractors of an antithetic integral feedback system have a simple structure

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Abstract—Since its introduction by Briat, Gupta and Khammash, the antithetic feedback controller design has attracted considerable attention in both theoretical and experimental systems biology. The case in which the plant is a two-dimensional linear system (making the closed-loop system a four-dimensional nonlinear system) has been analyzed in much detail. This system has a unique equilibrium e but, depending on parameters, it may exhibit periodic orbits. An interesting question is for what parameter values periodic orbits exist. Another open question is whether other dynamical behaviors, such as chaotic attractors, might be possible for some parameter choices. We show that, for any parameter choices, every compact omega-limit set that does not include e is a periodic solution. We also show that if the Jacobian of the vector field at the equilibrium is unstable then a (non-trivial) periodic orbit exists, and provide an explicit open set of initial conditions that are attracted to a periodic orbit. The analysis is based on the theory of strongly 2-cooperative systems.

I. INTRODUCTION

Feedback control is used by both natural and engineered systems, and in particular biological ones, in order to maintain the values of critical variables at precise homeostatic levels, or to force these variables to track reference signals so as to meet specifications [3]. Examples of such variables at the cellular level include concentrations of proteins and other chemical species; at an organism level, variables such as blood sugar, blood pressure, water content, or temperature are often finely regulated. Feedback attenuates the degradation of performance due to disturbances as well as uncertainties not accounted for when originally designing (or evolving) a controller.

Integral feedback (and more generally an “internal model” of the possible “exosignals”) is necessary for regulation, as is well known in linear systems theory as well as for certain classes of nonlinear systems, and plays a central role in biological processes, both at the organism and cellular level [9]. Recently, there has been a considerable effort directed at the construction of integral feedback controllers in synthetic biology (see [12] and the references therein). One particular approach that has gained much recent attention, and the focus of this work, relies on two-molecule sequestration or

annihilation reactions, and was introduced under the name “antithetic controllers” by Briat, Gupta, and Khammash [10]. Interestingly, the paper [6] shows mathematically that antithetic controllers are in a certain sense minimal-complexity controllers that achieve robust perfect adaptation.

Several research groups have produced implementations of the antithetic architecture. An *in vivo* construct was reported in [6] that is based on *Bacillus subtilis* sigW and its anti-factor rsiW as the two antithetic species. Another implementation [2] constructed and analyzed an *in vitro* synthetic biomolecular integral controller that precisely controls the protein production rate of an output gene, based on a sequestration reaction between *E. coli* σ_{28} and anti- σ_{28} factors.

Thus motivated, the purpose of this work is to analyze the following four-dimensional nonlinear dynamical system from [10]:

$$\begin{aligned}\dot{x}_1 &= \alpha_1 - \alpha_2 x_1 x_4, \\ \dot{x}_2 &= \alpha_3 x_1 - \alpha_4 x_2, \\ \dot{x}_3 &= \alpha_5 x_2 - \alpha_6 x_3, \\ \dot{x}_4 &= \alpha_7 x_3 - \alpha_8 x_1 x_4,\end{aligned}\tag{1}$$

with $\alpha_i > 0$. (In [10], $\alpha_2 = \alpha_8$, but we do not need that for the general theory.) We study this system in the nonnegative orthant $\mathbb{R}_+^4$, which is a (positively) invariant set of the dynamics and is the natural biological space to consider since these variables represent chemical species concentrations. This system is (with minor variations) also studied in [2], [1], where local stability and some partial global stability results are obtained (see also [11]).

The system (1) represents the closed-loop interconnection of an antithetic controller (described by the variables x_1 and x_4) and a simple two-dimensional linear system (described by the variables x_2 and x_3). It is easy to verify that this system has a unique equilibrium, for any (positive) parameter values. However, as shown in [10], a Hopf bifurcation may arise so that, depending on the parameters, the system can exhibit periodic orbits. An interesting open question is whether other dynamical behaviors, such as chaotic attractors, might be possible for some parameter choices.

Here, we show that (1) is a strongly 2-cooperative system. This has several implications. First, for any parameter choices, every bounded trajectory satisfies a strong Poincaré-Bendixson property [39]. Intuitively speaking, this suggests that every bounded solution that does not converge to the equilibrium converges to a periodic orbit. Second, we show

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that for any choice of parameter values such that the equilibrium is not locally asymptotically stable, the system admits a periodic solution and there exists a set of positive measure of initial conditions such that any solution emanating from this set converges to a (non-trivial) periodic orbit. Our results build on the theory of strongly 2-cooperative systems [39], [22]. However, these results typically assume that the system admits a bounded invariant set. For the system (1), we cannot prove the existence of such a set, so we need to modify the proofs. Furthermore, the special structure of (1) allows to derive some information on the average over a period of any periodic orbit.

The next section provides a literature review on k -cooperative systems that are a generalization of cooperative systems. Section III reviews known definitions and results that are used later on. Section IV details our main results, and their proofs are given in Section V.

II. K-COOPERATIVE SYSTEMS

We use the recently developed theory of k -cooperative dynamical systems [39] to analyze (1). Let P_-^k denote the set of vectors with no more than $k - 1$ sign variations (see the exact definition below). A linear time-varying dynamical system is called k -positive if its flow maps P_-^k to P_-^k , and strongly k -positive if its flow maps P_-^k to the interior of P_-^k . For $k = 1$, this reduces to the definition of a positive and strongly positive linear system [15].

A nonlinear system is called [strongly] k -cooperative if along any trajectory its associated variational system is [strongly] k -positive. For $k = 1$, this reduces to a [strongly] cooperative system [36]. An easy to verify sufficient condition for k -cooperativity is based on the k -additive compound matrix of the Jacobian of the vector field [7], [20]. For more applications of the k -additive compound, see [27], [24], [40].

The analysis of k -cooperative dynamical systems is based on the work of Sanchez [32] (see also the more recent work in [16], [17], [18]) on dynamical systems with invariant cones of rank k .

Definition 1: [23] A set $C \subset \mathbb{R}^n$ is called a cone of rank k if: (a) C is closed, (b) $x \in C$ implies that $ax \in C$ for all $a \in \mathbb{R}$, and (c) C contains a subspace of dimension k and no subspace of a higher dimension.

For example, it is straightforward to verify that $\mathbb{R}_+^2 \cup (-\mathbb{R}_+^2)$ (and more generally $\mathbb{R}_+^n \cup (-\mathbb{R}_+^n)$) is an invariant cone of rank 1. Note that this example shows that a cone of rank k is not necessarily convex.

Sanchez [32] studied dynamical systems for which the flow of the associated variational equation maps a cone of rank k to its interior. He showed that certain bounded trajectories (called pseudo-ordered trajectories) contained in an omega limit set can be projected in a one-to-one way to the invariant set of a Lipschitz dynamical system evolving on a k -dimensional space. He also showed that for $k = 2$ this implies that these trajectories satisfy a Poincaré-Bendixson property. Roughly speaking, this means that any such bounded pseudo-ordered trajectory that does not converge to an equilibrium converges to a periodic orbit.

It is not difficult to show that P_-^k is a cone of rank k . Thus, strongly k -cooperative dynamical systems can be analyzed using the theory developed in [32], [16]. However, the special nested structure of k -cooperative systems allows to deduce much more [39]. Roughly speaking, the results above hold for *every* bounded trajectory rather than just bounded pseudo-ordered trajectories. In particular, every bounded trajectory of a strongly 2-cooperative systems satisfies the strong Poincaré-Bendixson property. The recent paper [22] considered strongly 2-cooperative systems whose trajectories evolve on a bounded state space that includes a unique equilibrium point e , and derived a simple sufficient condition guaranteeing that some trajectories do not converge to e and thus converge to a periodic orbit.

Two additional advantages of k -cooperative systems are: (a) the projection can be described in a simple and explicit form using the eigenvectors of a suitable matrix; and (b) the condition for k -cooperativity is an easy to check *sign-pattern condition* on the Jacobian of the vector field.

Some related work includes the following. Linear mappings that preserve the number of sign variations in a vector have been studied for a long time in the context of the sign variations diminishing property of totally positive (TP) matrices, that is, matrices with all minors positive [19]. Schwarz [33] extended these ideas to dynamical linear systems. We call a matrix $A \in \mathbb{R}^{n \times n}$ Jacobi if A is tridiagonal with positive entries on the super- and sub-diagonals. Schwarz [33] proved that the transition matrix of $\dot{x} = Ax$, that is, $\exp(At)$ is TP for all $t > 0$ if and only if A is Jacobi. Such a system is called a totally positive differential system (TPDS).

Since $x(t) = \exp(A(t - t_0))x(t_0)$ for all $t \geq t_0$, the sign variations diminishing property implies that if A is Jacobi then the number of sign variations in $x(t)$ can only decrease with t . In the terminology of k -positive systems, this implies that the system is strongly k -positive for all $k \in \{1, \dots, n\}$.

In a seemingly different line of research, Smillie [34] studied the nonlinear dynamical system $\dot{x} = f(x)$ using the number of sign variations in the vector of derivatives $\dot{x}$ as an integer-valued Lyapunov function. He assumed that the Jacobian $J(x) := \frac{\partial f}{\partial x}(x)$ is a Jacobi matrix for all x . This was further developed by Smith [35] who considered time-varying and periodic nonlinear systems of a similar form. The recent paper [26] pointed out the connection between this work and the notion of a totally positive differential system (TPDS) introduced by Schwarz. Indeed, for $z(t) := \dot{x}(t)$, we have

$$\dot{z}(t) = J(x(t))z(t), \quad (2)$$

and if $J(x(t))$ is Jacobi for all t then (2) is a TPDS.

Mallet-Paret and Smith [25] studied a nonlinear system $\dot{x} = f(x)$ with a cyclic feedback structure, i.e. $\dot{x}_i = f_i(x_{i-1}, x_i)$, for $i = 1, \dots, n$, with x_0 interpreted as x_n . They showed that under certain assumptions on the signs of the nonzero entries of the Jacobian the number of *cyclic* sign variations in $\dot{x}$ is a Lyapunov function, and used this to prove a Poincaré-Bendixson property. Ref. [29] describes

an algorithm for mapping measurements of state-variables to a dynamical system with a cyclic feedback structure. The theory of linear dynamical systems that preserve the number of cyclic sign variations, called *cyclic totally positive differential systems* (CTPDSs), was further developed in [8]. Elkhader [13] generalized the results of Mallet-Paret and Smith to a system in the form:

$$\begin{aligned}\dot{x}_1 &= f_1(x_1, x_n), \\ \dot{x}_i &= f_i(x_{i-1}, x_i, x_{i+1}), \text{ for } i \in \{2, 3, \dots, n-1\}, \\ \dot{x}_n &= f_n(x_{n-1}, x_n),\end{aligned}$$

i.e. the dynamics for $\dot{x}_2, \dots, \dot{x}_n$ admits a tridiagonal structure (but not for $\dot{x}_1$), and there is a feedback connection from x_n to $\dot{x}_1$ (but not from x_1 to $\dot{x}_n$). Again, he assumed certain sign conditions on the entries of the Jacobian. However, the biological model studied here admits a more general structure and it seems that it cannot be analyzed using the results described above.

If $P \in \mathbb{R}^{n \times n}$ is symmetric with k $[n-k]$ negative [positive] eigenvalues then the set $C(P) := \{x \in \mathbb{R}^n \mid x^T P x \leq 0\}$ is a cone of rank k [23], [32]. This has been used to prove a Poincaré-Bendixson property for a certain type of dynamical systems in [37], [38]. A recent paper [28] studied the same biological model studied here using a similar technique, but only for certain specific parameter values. The analysis in [28] suggests that the model does not have a fixed $C(P)$ as an invariant set. Rather, for different parameter values it admits different invariant sets in the form $C(P)$. Discrete-time k -positive systems have been recently introduced and analyzed in [5], [21], [4].

III. PRELIMINARIES

We begin by reviewing several known results on the number of sign variations in a vector.

A. Sign variations

For $y \in \mathbb{R}^n \setminus \{0\}$, let $s^-(y)$ denote the number of sign variations in y after deleting all its zero entries. For example, for $n = 4$ and $z = [1 \ 0 \ 0 \ -1]^T$, $s^-(z) = 1$. Let $s^+(y)$ denote the maximal possible number of sign variations in y after replacing each zero entry with either 1 or -1 . For example, for $n = 4$ and $z = [1 \ 0 \ 0 \ -1]^T$, $s^+(z) = 3$. For the zero vector, define $s^-(0) := 0$ and $s^+(0) := n - 1$. Then

$$0 \leq s^-(y) \leq s^+(y) \leq n - 1 \text{ for all } y \in \mathbb{R}^n.$$

The following result provides information on the number of sign variations in the limit of a sequence of vectors.

Lemma 1: [30, Ch. 3] Let $v^k \in \mathbb{R}^n$, $k = 1, 2, \dots$, be a sequence such that $v = \lim_{k \rightarrow \infty} v^k$ exists. Then

$$s^-(v) \leq \liminf_{k \rightarrow \infty} s^-(v^k) \leq \limsup_{k \rightarrow \infty} s^+(v^k) \leq s^+(v). \quad (3)$$

The proof follows from the fact that if zero entries appear in the limit vector v this can only decrease [increase] s^- [s^+].

We can now define the sets of vectors with no more than $k - 1$ sign variations. For any $k \in \{1, \dots, n - 1\}$, let

$P_-^k := \{y \in \mathbb{R}^n \mid s^-(y) \leq k - 1\}$. Lemma 1 implies that P_-^k is a closed set. Furthermore, $P_+^k := \{y \in \mathbb{R}^n \mid s^+(y) \leq k - 1\}$ is the interior of P_-^k [39]. For example, $P_-^1 = \mathbb{R}_+^n \cup (-\mathbb{R}_+^n)$, and $P_+^1 = \text{int}(\mathbb{R}_+^n \cup (-\mathbb{R}_+^n))$. Note that this example shows in particular that the sets P_-^k and P_+^k are in general *not* convex sets. For a geometric analysis of these sets, see [39].

It is straightforward to see that $\mathbb{R}_+^2 \cup (-\mathbb{R}_+^2)$ (and, more generally, $\mathbb{R}_+^n \cup (-\mathbb{R}_+^n)$) is a cone of rank 1. Thus, cooperative systems admit an invariant cone of rank 1.

A cone C of rank k is called *solid* if its interior is nonempty, and *k-solid* if there is a linear subspace $\mathcal{W}$ of dimension k such that $\mathcal{W} \setminus \{0\} \subseteq \text{int}(C)$. In the context of dynamical systems, such cones are important because trajectories of dynamical systems that are confined to C can be projected to the linear subspace $\mathcal{W}$ [32]. Roughly speaking, if this projection is one-to-one then the trajectories must satisfy the same properties as trajectories in a k -dimensional space. It was shown in [39] that for any $k \in \{1, 2, \dots, n - 1\}$ the set P_-^k is a k -solid cone, and that its complement $\text{clos}(\mathbb{R}^n \setminus P_-^k)$ is an $(n - k)$ -solid cone.

B. Oscillatory matrices and induced linear subspaces

A matrix is called totally nonnegative (TN) if all its minors are nonnegative and totally positive (TP) if they are all positive. These matrices have many special and important properties [30], [14]. A matrix $A \in \mathbb{R}^{n \times n}$ is called *oscillatory* if it is TN and there exists an integer $\ell \geq 1$ such that A^ℓ is TP [19]. The smallest such ℓ is called the exponent of the oscillatory matrix A [41]. Oscillatory matrices have a special spectral structure.

Proposition 1: [19], [31] If $A \in \mathbb{R}^{n \times n}$ is an oscillatory matrix then its eigenvalues are all real, positive, and distinct. Order the eigenvalues as $\lambda_1 > \lambda_2 > \dots > \lambda_n > 0$, and let $u^k \in \mathbb{R}^n$ denote the eigenvector corresponding to λ_k . For any $1 \leq i \leq j \leq n$, let $\mathcal{V}_i^j := \text{span}(u^i, u^{i+1}, \dots, u^j)$. Then

$$i - 1 \leq s^-(v) \leq s^+(v) \leq j - 1 \text{ for any } v \in \mathcal{V}_i^j \setminus \{0\}. \quad (4)$$

Remark 1: Note that (4) implies in particular that $s^-(u^i) = s^+(u^i) = i - 1$ for all $i \in \{1, \dots, n\}$. Also, for any set in the form $P(k_1, k_2) := \{x \in \mathbb{R}^n \mid k_1 \leq s^-(x) \leq s^+(x) \leq k_2\}$, the eigenvectors of an oscillatory matrix can be used to provide an explicit representation of a subspace $\mathcal{W}(k_1, k_2)$ satisfying $\mathcal{W}(k_1, k_2) \setminus \{0\} \subseteq P(k_1, k_2)$.

IV. MAIN RESULTS

We now state our main results. The proofs are placed in Section V.

Proposition 2: The system (1), with $\alpha_i > 0$, admits a unique equilibrium e in $\mathbb{R}_+^4$. Fix an initial condition $x_0 \in \mathbb{R}_+^4$. If $x(t, x_0)$ is bounded and there exist $d > 0$ and $\tau > 0$ such that $|x(t, x_0) - e| \geq d$ for all $t \geq \tau$, then $x(t, x_0)$ converges to a periodic orbit.

Recall that $\omega(x_0)$, the omega limit set corresponding to x_0 , is the set of all z such that there exists a sequence $t_1 < t_2 < \dots$ with $t_i \rightarrow \infty$ and $\lim_{i \rightarrow \infty} x(t_i, x_0) = z$. If $x(t, x_0)$ is bounded then $\omega(x_0)$ is non-empty and compact, and Prop. 2 implies that if $e \notin \omega(x_0)$ then $\omega(x_0)$ is a periodic orbit. In other words, the attractor admits a simple structure.

Prop. 2 does not rule out the possibility that a solution $x(t, x_0)$ converges to a chaotic attractor, but in this case it guarantees that $e \in \omega(x_0)$. Prop. 2 also does not rule out the possibility that every bounded solution $x(t, x_0)$ converges to e , and then the system has no oscillatory behavior. The next result provides a simple sufficient condition guaranteeing that the system admits at least one (non-trivial) periodic orbit. Let $J(x)$ denote the Jacobian of the vector field in (1).

Proposition 3: Consider (1) with $\alpha_i > 0$. Let e denote its unique equilibrium in $\mathbb{R}_+^4$. Suppose that $J(e)$ admits at least one eigenvalue with a positive real part. Fix an initial condition $x_0 \in \mathbb{R}_+^4 \setminus \{e\}$ such that $s^-(x_0 - e) \leq 1$ and the solution $x(t, x_0)$ is bounded. Then $x(t, x_0)$ converges to a (non-trivial) periodic orbit.

This implies that bounded trajectories emanating from initial conditions x_0 that are “close” to e in terms of the number of sign variations in the vector $x_0 - e$ will not converge to e , but rather to a periodic orbit.

The special structure of the system (1) also allows to derive the following result.

Proposition 4: Let $z : [0, T) \rightarrow \mathbb{R}_+^4$ be a periodic orbit of the system, that is, $z(0) = z(T)$. For any T -periodic function y let $\bar{y} := \frac{1}{T} \int_0^T y(t) dt$, i.e., the average of y along a period. Then $\bar{z}_i = e_i$, $i = 1, 2, 3$, and $\bar{z}_4 \bar{z}_1 = e_4 \bar{z}_1$.

This suggests that any periodic solution “includes” the equilibrium point e .

Example 1: Consider (1) with $\alpha_1 = \alpha_3 = \alpha_6 = \alpha_7 = 1$, $\alpha_2 = \alpha_8 = 50$, $\alpha_4 = 3$, and $\alpha_5 = 14$. Then

$$e = [3/14 \quad 1/14 \quad 1 \quad 7/75]^\top \\ \approx [0.2143 \quad 0.0714 \quad 1 \quad 0.0933]^\top$$

(all numerical values are to four-digit accuracy). The eigenvalues of $J(e)$ are $\lambda_1 = 0.0044 + 1.5518i$, $\lambda_2 = 0.0044 - 1.5518i$, $\lambda_3 = -4.0643$, $\lambda_4 = -15.3255$. Fig. 1 depicts $x(t, x_0)$ with $x_0 = 0$. Note that $s^-(x_0 - e) = 0$. It may be seen that $x(t, x_0)$ does not converge to the equilibrium and converges to a periodic orbit z . Numerically integrating along the periodic solution gives $\bar{z}_1 = 0.2143$, $\bar{z}_2 = 0.0714$, $\bar{z}_3 = 0.9999$, $\bar{z}_4 = 0.0975$.

V. PROOFS OF MAIN RESULTS

1) Proof of Prop. 2: Clearly, (1) admits a unique equilibrium point

$$e = \left[\frac{\alpha_1 \alpha_4 \alpha_6 \alpha_8}{\alpha_2 \alpha_3 \alpha_5 \alpha_7} \quad \frac{\alpha_1 \alpha_6 \alpha_8}{\alpha_2 \alpha_5 \alpha_7} \quad \frac{\alpha_1 \alpha_8}{\alpha_2 \alpha_7} \quad \frac{\alpha_3 \alpha_5 \alpha_7}{\alpha_4 \alpha_6 \alpha_8} \right]^\top, \quad (5)$$

which is in the interior of $\mathbb{R}_+^4$. Recall that $\dot{x} = f(x)$ is 2-cooperative if its Jacobian $J(x) := \frac{\partial}{\partial x} f(x)$ satisfies the sign pattern

$$\begin{bmatrix} * & \geq 0 & 0 & \dots & 0 & \leq 0 \\ \geq 0 & * & \geq 0 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & * & \geq 0 \\ \leq 0 & 0 & 0 & \dots & \geq 0 & * \end{bmatrix}, \quad (6)$$

for every x in the state space [39]. Here $*$ denotes “don’t care”. It is strongly 2-cooperative if, in addition, $J(x)$ is an irreducible matrix almost everywhere. Consider the system (1) described in the introduction, analyzed on the nonnegative orthant $\mathbb{R}_+^4$, which is a forward invariant set for its dynamics. Its Jacobian is $J(x) =$

$$\begin{bmatrix} -\alpha_2 x_4 & 0 & 0 & -\alpha_2 x_1 \\ \alpha_3 & -\alpha_4 & 0 & 0 \\ 0 & \alpha_5 & -\alpha_6 & 0 \\ -\alpha_8 x_4 & 0 & \alpha_7 & -\alpha_8 x_1 \end{bmatrix}, \text{ and this satisfies the sign}$$

pattern (6). Furthermore, $J(x)$ is irreducible provided that $x_1 > 0$. Note that if at some time τ we have $x_1(\tau) = 0$ then (1) gives $\dot{x}_1(\tau) = \alpha_1 > 0$. This implies that $x_1(t) > 0$ for all $t > 0$, so $J(x(t))$ is irreducible for all $t > 0$. Using the strong Poincaré-Bendixson property of strongly 2-cooperative systems [39, Thm. 12] completes the proof of Prop. 2.

2) Proof of Prop. 3: We begin by presenting an auxiliary result that may be of independent interest and will be used later on. By a *cone* in $\mathbb{R}^n$ we mean a set $Q \subseteq \mathbb{R}^n$ with the property that for all $q \in Q$ and all $r > 0$, $rq \in Q$. Let $Q^c := \mathbb{R}^n \setminus Q$.

Lemma 2: Suppose that Q is an open cone and that there is a linear subspace $\mathcal{V} \subseteq \mathbb{R}^n$ such that $\mathcal{V} \setminus \{0\} \subseteq Q$. Pick any complement $\mathcal{U}$ of $\mathcal{V}$, i.e. a linear subspace such that $\mathbb{R}^n = \mathcal{U} \oplus \mathcal{V}$. Let $T : \mathbb{R}^n \rightarrow \mathcal{U}$ be the projection on $\mathcal{U}$ along $\mathcal{V}$. In other words, given any $x \in \mathbb{R}^n$ and its unique decomposition $x = u + v$ with $u \in \mathcal{U}$ and $v \in \mathcal{V}$, $Tx := u$. Then, there is some $c > 0$ such that $\|Tx\| \geq c\|x\|$ for all $x \in Q^c$.

Note that T is not required to be an orthogonal projection.

Proof: Assume on the contrary that there exists a sequence of elements $x^k \in Q^c$ and positive numbers $\varepsilon_k \rightarrow 0$ such that

$$\|Tx^k\| \leq \varepsilon_k \|x^k\|. \quad (7)$$

Let $x^k = u^k + v^k$, $u^k \in \mathcal{U}$, $v^k \in \mathcal{V}$. Then (7) gives $\|u^k\| \leq \varepsilon_k \|x^k\|$ for all k . Introduce

$$\tilde{x}^k := x^k / \|x^k\|, \quad \tilde{u}^k := u^k / \|x^k\|, \quad \tilde{v}^k := v^k / \|x^k\|$$

and note that $\|\tilde{u}^k\| \leq \varepsilon_k$, so $\tilde{u}^k \rightarrow 0$ as $k \rightarrow \infty$. Since every $\tilde{x}^k$ has unit norm, we may assume, taking a subsequence if necessary, that $\tilde{x}^k \rightarrow x^*$ as $k \rightarrow \infty$, for some $x^* \in \mathbb{R}^n$ of unit norm. Since $\tilde{v}^k = \tilde{x}^k - \tilde{u}^k$, $\tilde{x}^k \rightarrow x^*$, and $\tilde{u}^k \rightarrow 0$, it follows that $\tilde{v}^k \rightarrow x^*$ as $k \rightarrow \infty$. Since $v^k \in \mathcal{V}$ and $\mathcal{V}$ is a subspace, also $\tilde{v}^k \in \mathcal{V}$ for all k . Moreover, as $\mathcal{V}$ is closed (being a subspace) and $x^* \neq 0$ (it has unit norm) it follows that $x^* \in \mathcal{V} \setminus \{0\} \subseteq Q$. Since Q is open and $\tilde{x}^k \rightarrow x^*$, there is some k such that $\tilde{x}^k \in Q$. Since Q is a cone, also $x^k = \|x^k\| \tilde{x}^k \in Q$. This contradicts the assumption that $x^k \in Q^c$, and thus completes the proof of Lemma 2. ■

The next result demonstrates a corollary of Lemma 2 that will be used later on.

Corollary 1: Let $Q = \mathbb{R}^n \setminus P_+^k = \{x \in \mathbb{R}^n \mid s^+(x) \geq k\}$. This is an open cone, as $s^+(z) = s^+(rz)$ for any $r > 0$ (and also for any $r < 0$). Then $Q^c = \{x \in \mathbb{R}^n \mid s^+(x) \leq k-1\}$. Let $\mathcal{V}$ be a linear subspace of dimension $n-k$ such

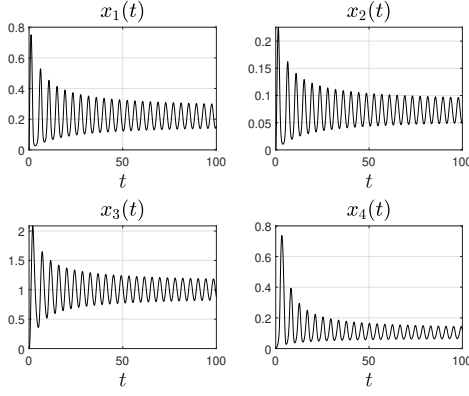


Fig. 1. State-variables of the system in Example 1 as a function of time.

that $\mathcal{V} \subseteq Q$, and let $\mathcal{U}$ be a linear subspace of dimension k such that $\mathcal{U} \setminus \{0\} \subseteq Q^c$ (see Remark 1). Then there exists $c > 0$ such that the following property holds. For any $x \in \mathbb{R}^n$ with $s^+(x) \leq k - 1$, let $x = u + v$ with $u \in \mathcal{U}$ and $v \in \mathcal{V}$. Then $\|x\| \leq \|u\|/c$.

Intuitively speaking, this implies that decomposing any vector x that has a small number of sign variations as $x = u + v$, where $u \in \mathcal{U}$ [$v \in \mathcal{V}$] has a small [large] number of sign variations then $\|v\|$ will always be small w.r.t. to $\|x\|$.

Ref. [22] considers a strongly 2-cooperative system $\dot{x} = f(x)$, with $f \in C^2$, and with an open, bounded and convex forward invariant state space $\Omega \subset \mathbb{R}^n$, that includes a unique equilibrium e . It is shown that if $J_f(e) := \frac{\partial}{\partial x} f(x)|_{x=e}$ has at least two eigenvalues with a positive real part then for any $x_0 \in \Omega \setminus \{e\}$ with $s^-(x_0 - e) \leq 1$ the trajectory $x(t, x_0)$ does not converge to e and thus the boundness of Ω and the strong Poincaré-Bendixson property implies that it converges to a (non-trivial) periodic orbit as $t \rightarrow \infty$.

However, for the system (1), we cannot prove the existence of a bounded forward invariant state-space, so we need to modify the proof. In fact, we provide a more direct and transparent proof using the specific structure of (1).

Let $p(t) := x(t) - e$. Then

$$\begin{aligned} \dot{p}_1 &= -\alpha_2(p_1 p_4 + e_1 p_4 + e_4 p_1), \\ \dot{p}_2 &= \alpha_3 p_1 - \alpha_4 p_2, \\ \dot{p}_3 &= \alpha_5 p_2 - \alpha_6 p_3, \\ \dot{p}_4 &= \alpha_7 p_3 - \alpha_8(p_1 p_4 + e_1 p_4 + e_4 p_1). \end{aligned} \quad (8)$$

This system admits a unique equilibrium at zero. Its Jacobian

at zero is $J_p(0) = \begin{bmatrix} -\alpha_2 e_4 & 0 & 0 & -\alpha_2 e_1 \\ \alpha_3 & -\alpha_4 & 0 & 0 \\ 0 & \alpha_5 & -\alpha_6 & 0 \\ -\alpha_8 e_4 & 0 & \alpha_7 & -\alpha_8 e_1 \end{bmatrix}$. The

next result describes some useful spectral properties of $J_p(0)$.

Lemma 3: Let $\lambda_i \in \mathbb{C}$, $i = 1, \dots, 4$, denote the eigenvalues of $J_p(0)$ ordered such that complex conjugate eigenvalues appear consecutively, and

$$\operatorname{Re}(\lambda_1) \geq \operatorname{Re}(\lambda_2) \geq \operatorname{Re}(\lambda_3) \geq \operatorname{Re}(\lambda_4). \quad (9)$$

Suppose that at least one eigenvalue has a positive real part. Then

$$\lambda_1 = u + iv, \lambda_2 = u - iv, \text{ with } u > 0, v \neq 0, \quad (10)$$

$\operatorname{Re}(\lambda_3) < u$, and $\operatorname{Re}(\lambda_3) + \operatorname{Re}(\lambda_4) < -2u$.

Note that this implies in particular that the equilibrium e is a saddle point.

Proof: A calculation gives $\det(sI_4 - J_p(0)) = s^4 + \beta_3 s^3 + \beta_2 s^2 + \beta_1 s + \beta_0$, with $\beta_3 := \alpha_4 + \alpha_6 + \alpha_2 e_4 + \alpha_8 e_1$, $\beta_2 := \alpha_4 \alpha_6 + \alpha_2 \alpha_4 e_4 + \alpha_2 \alpha_6 e_4 + \alpha_4 \alpha_8 e_1 + \alpha_6 \alpha_8 e_1$, $\beta_1 := \alpha_2 \alpha_4 \alpha_6 e_4 + \alpha_4 \alpha_6 \alpha_8 e_1$, and $\beta_0 := \alpha_2 \alpha_3 \alpha_5 \alpha_7 e_1$. All the β_i s are positive, so $J_p(0)$ cannot have a real and non-negative eigenvalue. Since we assume that $J_p(0)$ admits at least one eigenvalue with a positive real part, the ordering (9) yields $\operatorname{Re}(\lambda_1) > 0$ and thus $\lambda_1 = u + iv$, with $u > 0$ and $v \neq 0$. By the ordering of the eigenvalues, $\lambda_2 = u - iv$.

Let $J_p^{[2]}(0)$ denote the 2-additive compound of $J_p(0)$. Then $J_p^{[2]}(0) \in \mathbb{R}^{6 \times 6}$, and its eigenvalues ζ_i , $i = 1, \dots, 6$, are all the sums $\lambda_i + \lambda_j$, with $i < j$. Order these in lexicographic order, i.e. $\zeta_1 = \lambda_1 + \lambda_2$, $\zeta_2 = \lambda_1 + \lambda_3$, $\dots$, $\zeta_6 = \lambda_3 + \lambda_4$. Then $\operatorname{Re}(\zeta_1) \geq \operatorname{Re}(\zeta_2) \geq \dots \geq \operatorname{Re}(\zeta_6)$. Since $J_p^{[2]}(0)$ is Metzler and irreducible, $\operatorname{Re}(\zeta_1) > \operatorname{Re}(\zeta_2)$, so $\operatorname{Re}(\lambda_2) > \operatorname{Re}(\lambda_3)$.

The characteristic polynomial of $J_p(0)$ is thus $\det(sI_4 - J_p(0)) = (s - (u + iv))(s - (u - iv))(s - \lambda_3)(s - \lambda_4)$. The coefficient of s^3 in this polynomial is $-\lambda_3 - \lambda_4 - 2u$. This is equal to $\beta_3 > 0$ and this completes the proof. ■

Remark 2: Let $v^1 + iv^2$, with $v^1, v^2 \in \mathbb{R}^4$, denote the eigenvector of $J_p(0)$ corresponding to the eigenvalue $\lambda_1 = u + iv$. Then the results in [4] imply that $c_1 v^1 + c_2 v^2 \in P_+^2$ for any $c_1, c_2 \in \mathbb{R}$ with $c_1^2 + c_2^2 \neq 0$.

Let $f_p : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ denote the vector field in (8). Let $T \in \mathbb{R}^{4 \times 4}$ be a non-singular matrix, and define $q(t) := Tp(t)$. Then $\dot{q} = Tf_p(T^{-1}q)$, and the Jacobian of this system at the equilibrium 0 is $J_q(0) = TJ_p(0)T^{-1}$. Combining Lemma 3 and [22, Prop. 3] implies that there exists a non-singular matrix T such that $J_q(0) = \begin{bmatrix} \Lambda & 0_{2 \times 2} \\ 0_{2 \times 2} & \Psi \end{bmatrix}$, where $\Lambda = \begin{bmatrix} u & -v \\ v & u \end{bmatrix}$, with $u > 0$, $v \neq 0$, and $\Psi \in \mathbb{R}^{2 \times 2}$. Specifically, the first and second columns of T^{-1} are v^1 and v^2 , respectively. Thus, $\dot{q} = \begin{bmatrix} \Lambda & 0_{2 \times 2} \\ 0_{2 \times 2} & \Psi \end{bmatrix} q + o(q)$, where $o(q)$ represents terms that are at least quadratic in $q_1, \dots, q_4$.

Define $V : \mathbb{R}^4 \rightarrow \mathbb{R}_+$ by $V(z) := (z_1^2 + z_2^2)/2$. Then along any solution of the q -system, we have

$$\begin{aligned} \dot{V}(q) &= q_1 \dot{q}_1 + q_2 \dot{q}_2 \\ &= \begin{bmatrix} q_1 & q_2 \end{bmatrix} \begin{bmatrix} u & -v \\ v & u \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} + o(q^2) \\ &= u(q_1^2 + q_2^2) + o(q^2), \end{aligned} \quad (11)$$

where $o(q^2)$ represents terms that are at least cubic in $q_1, \dots, q_4$.

Fix an initial condition $a \neq 0$ such that $s^-(a - 0) \leq 1$ (recall that 0 is an equilibrium of the q system, i.e., $q(t, 0) =$

0 for all $t \geq 0$) and $q(t, a)$ is bounded. Since the system is strongly 2-cooperative, $s^+(q(t, a) - q(t, 0)) \leq 1$ for all $t > 0$. Combining Remark 2 and Corollary 1 implies that there exists $c > 0$ such that $\|q(t, a)\| \leq c^{-1} \left\| \begin{bmatrix} q_1(t, a) \\ q_2(t, a) \end{bmatrix} \right\|$ for all $t > 0$. Combining this with (11) and the fact that $u > 0$ implies that there exists $\eta > 0$ such that $\dot{V}(q(t, a)) \geq 0$ whenever $q_1^2(t, a) + q_2^2(t, a) \leq \eta$. Thus, $q(t, a)$ keeps a positive distance from the unique equilibrium 0. Using the strong Poincaré-Bendixson property completes the proof of Prop. 3.

Example 2: Reconsider Example 1. The eigenvector corresponding to λ_1 is $v^1 + iv^2$, with $v^1 = [-0.0388 \ -0.0639 \ -0.8913 \ -0.0404]^\top$, $v^2 = [-0.3960 \ -0.0988 \ 0.0000 \ 0.1783]^\top$. As expected, $s^-(v^1) = 0$ and $s^-(v^2) = 1$. The eigenvectors corresponding to the real eigenvalues λ_3 and λ_4 are $v^3 = [-0.2219 \ 0.2085 \ -0.9524 \ 0.0125]^\top$, $v^4 = [0.7067 \ -0.0573 \ 0.0560 \ 0.7030]^\top$.

3) *Proof of Prop. 4:* Integrating (1) along a periodic orbit z yields: $\alpha_1 = \alpha_2 \bar{z}_1 \bar{z}_4$, $\alpha_3 \bar{z}_1 = \alpha_4 \bar{z}_2$, $\alpha_5 \bar{z}_2 = \alpha_6 \bar{z}_3$, $\alpha_7 \bar{z}_3 = \alpha_8 \bar{z}_1 \bar{z}_4$. Using this and (5) completes the proof.

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